Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/05/2007
PI Name:	Vandenburgh, Herman Ph.D.		
Project Title:	Effect of Space Travel on Skeletal Myofibers		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle		
Flight Assignment:	NOTE: Received NCE through 6/30/2007, per A. Chu (9/06) renewed. Updated Aug. 26, 2005.		
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Task Description:	Long-term manned space flight requires a better understanding of skeletal muscle atrophy which results from microgravity. Astronaut strength and dexterity must be maintained for normal mission operations, extravehicular activities, and emergency situations. An understanding at the cellular level of how gravity/tension/exercise regulate muscle size by altering protein synthesis and/or protein degradation rates will allow pharmacological intervention to attenuate or prevent muscle atrophy in space. Avian skeletal myofibers in organ-like structures (organoids) have been flown on two Shuttle Missions (STS66 and STS77) in the Space Tissue Loss (STL) Module and validated as a good cellular model to study countermeasures for space travel-induced muscle atrophy.The overall hypothesis of this proposal is that recombinant human growth hormone (rhGH) can significantly attenuate microgravity-induced muscle atrophy when provided from a cell based delivery system. Cell based delivery of growth factors such as rhGH is based on our ability to insert the genes for these proteins into tissue cultured cells, and then implant the cells in vivo as a retrievable living device secreting physiological levels of the factors. We have found in ground-based studies that cell based delivery of rhGH, most likely through its elevation of insulin-like growth factor-1 (IGF-1), is more effective than daily rhGH injections in attenuating hindlimb unloaded soleus muscle atrophy. In the first series of new Shuttle experiments, the middeck Cell Culture Module (CCM) or Cell Culture Unit (CCU) will be used to determine if cell based delivery of IGF-1 attenuates space travel-induced avian skeletal muscle atrophy in vitro. These studies will be followed by determining whether cell based delivery of rhGH attenuates skeletal muscle atrophy in rodents flown in the Animal Environmental Module (AEM) of the Shuttle. The results of these studies will directly test the use of cell based delivery of rhGH as a countermeasure for skeletal muscle atrophy in space. The technology also have numerous applications for the treatment of Earth based musculoskeletal disorders. The overall hypothesis of this proposal is that recombinant human growth hormone (rhGH) can significantly attenuate microgravity-induced muscle atrophy when provided from a cell based delivery system. The results of these studies will directly test the use of cell based delivery of rhGH as a countermeasure for skeletal muscle atrophy in space. The technology also have numerous applications for the treatment of Earth based musculoskeletal disorders.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While the primary goal of this project is to understand and treat space travel-induced skeletal muscle atrophy, the results from these studies may have applications for several skeletal muscle wasting disorders on Earth. These include the severe muscle wasting observed in paralyzed patients and in the frail elderly, both of which partially respond to the increased tension associated with exercise and physical therapy. By better understanding the interactions of microgravity and muscle atrophy, physical therapy could be optimized for increased patient mobility and independence.
	FINAL NASA RESEARCH REPORT SUMMARY I. Paracrine release of insulin-like growth factor 1 from a bioengineered tissue stimulates skeletal muscle growth in vitro. Results of these studies were reported in last year's progress report and have now been published in the peer-reviewed journal Tissue Engineering (Shansky et al., 2006) II. Optimization of Bioartificial Muscle Tissue Engineering A. Extracellular Matrix Adult skeletal muscle progenitor cells can be embedded in an extracellular matrix (ECM) and tissue-engineered to form BioArtificial Muscles (BAMs), composed of aligned postmitotic myofibers. A collagen type 1 (Zyderm) extra cellular matrix (ECM) has been previously employed, however, these collagen BAMs have shown anti-angiogenic properties, which may limit myofiber survival in vivo. Vascularization is important for the supply of nutrients, removal of wastes, and offloading of therapeutic proteins in vivo. Hyaluronan and fibrin were investigated as an alternative ECM for the engineering of BAMs. BAMs engineered in hyaluronan ECM did not contract to form organized myoblasts under uniform, uniaxial passive tension. BAMs were successfully engineered in a fibrin matrix. However, fibrin undergoes degradation in vivo in a matter of days due to fibrinolysis. To overcome subsequent structural degradation of the BAMs, in vitro culture conditions were optimized to enhance the deposition of de novo synthesized ECM by the addition of proline, hydroxyproline and ascorbic acid. The results of these studies will be utilized to engineer BAMs with more native ECM and greater survival potential in vivo. B. Viscoelastic Characterization Bio-Artificial Muscles (BAMs) were tissue engineered in vitro with specialized molds inducing passive tension throughout muscle growth. Muscles were made with varying proliferating myogenic progenitor cell numbers, extracellular matrix types and concentrations, media supplements, and culture times. Tensile testing of the BAMs was carried out with hardware that stretched the muscle at a fixed rate while measuring force according to a LabView software virtual instrument (VI). Muscles with both collagen type I and fibrinogen extracellular matrices (ECMs) contracted into constructs with fused myofibers displaying unidirectional alignment that showed a potential for stiffness (elastic modulus) values which optimize muscle striation. The viscoelastic properties of the fibrin BAMs and deposition of collagen were characterized. BAMs engineered with the addition of proline, hydroxyproline and ascorbic acid had an increased Young's Modulus, decreased maximum strain, and increased collagen deposition. Lowering the fibrin content of the BAMs also increased Young's Modulus, decreased maximum strain, and increased collagen deposition. Allowing for a degradation of the fibrin matrix during the in vitro culture period had a significant effect on the deposition of collagen in the presence of supplements. Consistent stress-strain curves for the bioengineered muscles validate the continued use of this testing system for future studies of passive mechanics in muscle tissue engineering. These studies will help to optimize the mechanical properties of the BAMs for in vivo implantation. B. Innervation and Myosin Heavy Chain Isoform Expression Adult skeletal muscle progenitor cells can be embedded in extracellular matrices (ECM) and differentiated into organized bundles of post mitotic myofibers (Bioartifical Muscles, BAMs). BAMs can potentially be used for implantation to restore the function of damaged muscle tissue, or genetically engineered to secrete therapeutic proteins. In order to compare the BAMs to the real biological tissue, we characterized the BAMs in terms of myosin heavy chain (MHC) expression. Myosin is a major component of the contractile apparatus of muscle with different isoforms

Task Progress:	expressed in a developmentally-regulated manner in skeletal muscle. We performed QPCR to investigate expression of MHC isoforms in BAM samples and 2-D cell cultures, and determine the degree of maturation of BAMs. After 6 days in culture, we found that the BAMs were at the perinatal stage in terms of MHC gene expression. They may have the potential to reach a more advanced developmental stage when further differentiated. BAMs can potentially be implanted intramuscularly and become innervated. We have been able to increase the number of neuromuscular-like acetylcholine receptor (AchR) clusters induced on muscle fibers by incubation with the AchR binding protein agrin and formation of the advanced structure- pretzel-shaped AchR clusters by treatment with the extracellular matrix protein laminin. These features offer a structural advantage for the purpose of innervation of the BAMs. As a result, the in-growth of host neurites may provide trophic support to the BAMs by forming neuromuscular junctions, up-regulating expression of growth factors, improving the survival and functionality of the BAMs, and further providing support to the biological integrity of the BAM. Before implanting the BAMs intramuscularly for muscle repair, we first implanted BAMs formed from GFP labelled cells subcutaneously to examine cell viability and potential host responses. We found that cell viability was high at Day7 and Day14 but decreased over the next 30 days as shown by fluorescent microscope images. This might be attributed to inflammatory responses and limited nutrient supply for the implant. Some signs of neovascularization were detected on Day14. C. Viral Transduction for Recombinant Protein Secretion We have continued to work on increasing gene transfer efficiency into skeletal muscle cells. AAV vectors and pseudotyped lentiviral vectors were purchased from the University of Pennsylvania Vector Core Labs and tested for their ability to transduce muscle cells. Multiplicity of infection was 10, 100 and 1000 infectious units (lentiviral) or genome copies (AAV) per cell. Percentage of transduced cells was monitored over a period of 23 days by fluorescence microscopy and FACS. Results indicate that none of the AAV serotypes is suitable for a stable myoblast transduction, most likely because the majority of AAV genomes are not inserted in the host genome and become diluted out as the myoblasts divide. In contrast, all lentiviral vectors did result in a stable transduction, with the VSV-G envelope protein the most effective for transduction of the myoblasts. With this pseudotype, after 23 days 72, 26 or 2 % of the cells were GFP positive following a single transduction with an MOI of 1000, 100 or 10 respectively. III. Improved Bioartificial Muscle Implantation on Resorbable Scaffolds Successful transplantation of skeletal muscle progenitor cells (satellite cells or myoblasts) for either structural repair or ex vivo gene therapy applications will require the optimization of cell survival and differentiation into long-lived postmitotic muscle fibers. Three dimensional biodegradable polymer scaffolds have been successfully utilized for growth and transplantation of numerous cell types, including rodent skeletal muscle cells, but have normally utilized non-clinical grade carrier proteins such as MatrigelTM to optimize cell retention and attachment to the porous scaffold material. The retention, growth and differentiation of primary adult human skeletal myoblasts on polymeric scaffolds was achieved utilizing clinical grade collagen or fibrin carrier proteins. Porous polymer scaffolds were fabricated from poly(lactide-co-glycolide) (PLG) using high-pressure carbon dioxide. Scaffolds 4 to 13 mm in diameter by 1.5 mm thick were fabricated from particulate PLG and 10% - 100% (wt/wt) microsphere PLG. Cells (1.5 - 16 million/scaffold) seeded on the scaffold without a carrier protein were poorly retained whereas the use of collagen type I (ZydermTM) or fibrin (TisseelTM) during seeding led to the retention of greater than 85% of the cells within the scaffold's pores. Cell viability in vitro for 3-4 days in growth medium was best on the surface of the scaffold, with decreasing viability from the surface to their center. When differentiation medium was added to the cell-seeded scaffolds for an additional 3-5 days in vitro, short isolated postmitotic muscle fibers expressing sarcomeric tropomyosin formed throughout the scaffold pores. Prefilling the scaffolds with a cell-free (ZydermTM) or fibrin (TisseelTM) carrier protein prior to plating a layer of cells in carrier protein onto the scaffold surface resulted in the formation of a 100 µm cell layer of well formed postmitotic muscle fibers. When implanted subcutaneously into immunodeficient mice, the muscle fibers were maintained under tension in vivo. Primary human skeletal myoblasts stably transduced with GFP were used to track muscle fiber survival in vivo. Preliminary data show a 60% cell survival at 2 weeks and 24% at 4 weeks post-implantation, which compares favorably to the use of cells in a non-tensioned carrier protein matrix alone (22% at 2 weeks, 5% at 4 weeks). Although immunodeficient animals were used, loss of cells may in part be attributable to the implantation of xenogeneic cells since inhibition of natural killer (NK) cells improved muscle fiber survival. Polymer scaffolds therefore provide a reliable method for the successful transfer of ex vivo differentiated postmitotic muscle fibers under tension in vivo for their potential future use in muscle repair or gene therapy applications. (Thorrez et al., 2008)

Bibliography Type:	Description: (Last Updated: 10/03/2014)
Articles in Peer-reviewed Journals	Shansky J, Creswick B, Lee P, Wang X, Vandenburgh H. "Paracrine release of insulin-like growth factor 1 from a bioengineered tissue stimulates skeletal muscle growth in vitro. " Tissue Eng. 2006 Jul;12(7):1833-41. PMID: 16889513 , Jul-2006
Articles in Peer-reviewed Journals	Thorrez L, Shansky J, Wang L, Fast L, Vandendriessche T, Chuah M, Mooney D, Vandenburgh H. "Growth, differentiation, transplantation and survival of human skeletal myofibers on biodegradable scaffolds." Biomaterials. 2008 Jan;29(1):75-84. PMID: 17928049 , Jan-2008
Articles in Peer-reviewed Journals	Lee PH, Vandenburgh H. " Spaceflight reduces foreign protein expression in tissue-engineered skeletal muscle." Gravit Space Biol. 2012 Apr;26(1):17-24. http://gravitationalandspacebiology.org/index.php/journal/article/view/563 ; accessed 9/30/14 , Apr-2012